

Assessing vaccination coverage and antibody levels for measles, pertussis, and tetanus in Boende, DR Congo: Implications for maternal and child health

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ABSTRACT

Introduction: Despite vaccines' global success, vaccine-preventable diseases/infections remain a major cause of child mortality. The DRC, and particularly Tshuapa province, faces persistent challenges in improving coverage. Most vaccination coverage studies in the region rely on vaccination cards and caregiver recall, which can be prone to inaccuracies.

Methods: We conducted a 30 × 7 cluster survey with a serology component in Boende Health Zone, Tshuapa. Antibodies against measles, pertussis, and tetanus were measured using ELISA on dried blood spots from postpartum women and children aged 12–23 months. Two measles cut-offs (150 and 275 IU/L) were applied due to lack of standardization.

Results: The study enrolled 205 children and 208 women. Among women, 73 % had tetanus antibody levels above the protective threshold, 0 % had antibodies against pertussis above the predetermined cut-off, and measles seropositivity was 55 % at the more stringent cut-off (275 IU/L) and 81 % at a more relaxed one (150 IU/L). In children, 18 % had tetanus antibodies above the protective threshold and 5 % had antibodies above the pertussis cut-off point. and 18 % and 30 % for measles (strict and relaxed, respectively). In five clusters, no children had antibodies above the cut-off points for any of the three pathogens.

Conclusion: These findings highlight the potential need to include pertussis vaccination in the national maternal immunization program and advocate for integrating serology into vaccination coverage surveys to provide more accurate assessments. The disparity between reported vaccination rates and serological data could be due to one or a combination of recall bias, desirability bias, test variability, antibody waning, and possible concerns about vaccine potency and record accuracy in remote areas in the Boende Health Zone. This stresses the urgent need to improve vaccination coverage and surveillance efforts in Boende Health Zone and the broader Tshuapa province.

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1. Introduction

Vaccine-Preventable Diseases (VPDs)² continue to be a public health threat worldwide, especially in some Low and Middle-Income Countries (LMICs) [1]. The Expanded Program on Immunization (EPI), has been instrumental in combating VPDs by guiding national vaccination efforts, often ensuring free access to vaccines through governmental and international funding mechanisms [2]. However, vaccination coverage is still not satisfactory in many countries. In 2023, 14.5 million children had not received any vaccine dose (so-called zero-dose children), and close to 6 million had started but not completed their vaccination schedule [3].

The Democratic Republic of the Congo (DRC) is one of the countries with low full childhood vaccination levels, 21 % according to the National Demographic and Health Survey (DHS) 2023–2024 data, and one of the highest mortality rates in children under the age of five years old (76 per 1000 live births) in 2022 [4,5]. Although vaccination coverage has increased in the past decade, the COVID-19 pandemic impacted this progress, affecting vaccination efforts, and funding, and eroding public trust [6–8]. The barriers to achieving optimal coverage in the DRC are multifactorial, including logistical challenges like poor road infrastructure and cold chain challenges, vaccine hesitancy, illicit practices and conflict in certain regions [9–11]. The low vaccination coverage has resulted in multiple outbreaks of VPDs across the country. For example, in 2023, WHO/UNICEF reported approximately 650,000 measles cases globally, with nearly half of these cases recorded in the DRC [12]. That year, the country experienced a widespread outbreak that caused nearly 6000 deaths [12].

The Tshuapa province in northwestern DRC has some of the lowest vaccination rates in the country. According to the preliminary report of the 2023–2024 DHS, only 1 % of children aged 12–23 months in Tshuapa were fully vaccinated³ [13].

There are multiple studies describing vaccination coverage in the DRC as suboptimal, but the majority of these estimates are based on parent recall or vaccination cards [14–16]. While these methods are practical for estimating coverage, they are prone to inaccuracies due to recall bias and/or the loss of vaccination cards. Incorporating a serological component allows for a more direct assessment of population immunity, providing insight into overall susceptibility and protection levels within the community [17]. Serological data could also offer a reliable basis for shaping national immunization policies, supporting more targeted interventions and tailored strategies, particularly in underserved areas like Tshuapa. While seropositivity may result from either vaccination or past infection, this distinction can often be clarified through a comprehensive survey of disease and vaccination history. Moreover, the presence of antibodies in individuals who do not cite being vaccinated may also help identify past outbreaks.

To our knowledge, the last published serology-based coverage data in the DRC dates from the 2013–2014 DHS. In 2023–2024 DHS conducted another serosurvey which had a small sample size, including only 126 children from Tshuapa, a province with 12 health zones⁴ and over a

million and half inhabitants [18,19]. However, the results have not been published yet, at the time of writing this manuscript. Therefore, we conducted a serosurvey in one health zone in Tshuapa province (Boende Health Zone of 364,649 inhabitants) to provide a more detailed mapping of the vaccination situation in the area. Similar to DHS studies, the target population was chosen to be children aged 12–23 months and women who recently gave birth in order to assess maternal and childhood vaccination coverage. The aim of this study was to shed light on the vaccination status and background immunity against measles, pertussis and tetanus in Boende Health Zone in a way that could inform policy to improve vaccine distribution in the area.

2. Methods

2.1. Study setting

Boende Health Zone is located in Tshuapa province, DRC (Fig. 1) [20]. Boende town, the provincial capital is considered remote -much like the entire province-as it is only accessible by weekly flights or by week-long boat rides from the capital Kinshasa. In 2019, according to data obtained from the Central Office of the Health Zone in Boende, the number of people living in Boende Health Zone was estimated at 364,649, which was the population count used in study.

Out of the antigens included in the DRC's national vaccination calendar (Supplementary Table 1), we focused on tetanus, measles, and pertussis. In the EPI schedule, tetanus and pertussis are administered together as part of the pentavalent DTP + HepB+Hib vaccine at 6 weeks, 10 weeks, and 14 weeks of age [21]. At the time of conducting the study, only one dose of a measles-containing vaccine was given at 9 months of age, which marked the completion of the EPI vaccination schedule for infants. As for maternal vaccination, only the tetanus and diphtheria vaccines are included in the vaccination program. The first two doses (Td1 and Td2) are administered during the first pregnancy or at first contact with the health system and for women of childbearing age during vaccination campaigns, with Td2 given at least four weeks after Td1. Three additional doses (Td3, Td4, and Td5) should be administered at specific intervals; Td3 six months to one year after Td2 (or in a subsequent pregnancy), Td4 one to five years after Td3, and Td5 one to ten years after Td4. This schedule ensures long-term protection, especially



Fig. 1. A map of DRC showing Tshuapa province and Boende Health Zone (June 2023) [20].

² While the majority of references in the literature continue to use the term Vaccine-Preventable Diseases (VPD), we acknowledge a growing trend to use the term Vaccine-Preventable Infections (VPI), particularly in immunological studies. In this manuscript, we use both terms and particularly VPI when referring to antibody responses, as these responses are typically directed against the infection rather than the clinical manifestation of disease.

³ The DHS's definition of "fully vaccinated" was to be vaccinated with BCG, three doses of the pentavalent DTP + HepB+Hib, three doses of polio vaccine (not including the polio dose at birth), and one dose of the measles vaccine.

⁴ A Health Zone is a well-defined geographic and administrative health district within a province, designed based on fixed demographic and geographic criteria. It serves at least 100,000 rural or 200,000 urban inhabitants through a two-tiered health system, consisting of primary health centers and a General Referral Hospital (HGR), supervised by a Health Zone Management Team [62].

during future pregnancies [22].

2.2. Sampling, participant recruitment, and data collection

The WHO's EPI coverage survey training module (WHO/IVB/08.07) was followed for sampling and the survey was conducted in June 2023 [23]. Briefly, and using these guidelines, we selected 30 out of 490 villages or neighborhoods, that were considered as 30 separate clusters. Then, we collected data from seven women who had given birth 0–11 months prior to data collection and seven children aged 12–23 months in each cluster, given that the DRC's vaccine schedule for full immunization was considered complete by 9 months at that time. The EPI 30×7 design was chosen due to its suitability for remote, resource-limited settings. The target sample size was $30 \times 7 = 210$ children and 210 women. More details on the sampling method can be found in the WHO guidance book [23].

Blood samples were collected using Dried Blood Spots (DBS). While serum is considered the gold standard for measles antibody detection, both WHO and CDC confirm that DBS can be used to analyze measles IgG in conditions where serum usage is not feasible [24,25]. DBS sampling was selected over venous blood collection due to its practicality in remote areas where overnight stays were required. Additionally, the minimal invasiveness and blood quantity (50 μ l (μ L) of spiked whole blood) collected in this approach may have mitigated worries of “blood selling” that were voiced during a previous qualitative study that our team conducted in Boende town [26]. In this study, DBS were collected from the capillary blood of fingers on Whatman® 903 Protein Saver Cards (Cytiva, product no. WHA10531018, Marlborough, USA). A nurse/doctor conducted the finger pricking to collect at least four blood spots from each participant.

Five teams, each composed of two local health workers from Boende, were recruited and trained to conduct the survey and DBS collection. Upon approaching a household, interviewers introduced themselves and the research objective and asked if there were any residents who fit the inclusion criteria. If a potential participant was identified, the woman or the child caregiver (the mother in most cases) received an explanation of the study and an Informed Consent Form (ICF) that was available in French and Lingala, the local language. After accepting, participants would sign/give a thumbprint over two copies of the ICF (one for the investigator and one for the participant). Afterward, the interviewer would start asking the survey questions. As compensation for their time, participants received small amounts of sugar, soap, and salt, and children received a piece of candy to ease the potential discomfort of blood sampling. If an eligible household declined participation or no eligible respondent was present at the time of the visit, the household was replaced by the nearest neighbouring household, as recommended in the WHO EPI cluster survey guidelines. No households were revisited. This substitution procedure is commonly used in operational cluster surveys but may introduce selection bias if absentees or refusals differ systematically from participating households.

After collection, DBS cards were left to dry in one house in the neighborhood/village with mesh to protect them from insects (Fig. 2). At the end of the day, each DBS card was placed in a Ziplock bag with a silica sachet (RS PRO, Silica Gel, 2 g, RS stockno.:388–8421) to absorb humidity. Once data collection was completed, DBS cards were shipped to Kinshasa and put in a -20°C freezer until they were shipped to Sciensano, Belgium for analysis.

The survey included sociodemographic questions in addition to exploring the vaccination history of the participant and children's disease history for measles. To assess economic status, we used access to amenities like electricity and ownership of devices (e.g., radio, mobile phone) as a proxy. Study data was collected and managed using REDCap electronic data capture tools hosted at the University of Antwerp [27,28]. The REDCap mobile app was installed on tablets; the interviewers were trained to use the mobile app and conduct the survey. The French version of the survey questions is provided in supplementary



Fig. 2. Dried Blood Spots (DBSs) collected during the survey and put in a children's mosquito net for protection (June 2023).

materials (Supplementary materials 1). The results regarding the coverage of polio and BCG vaccines were can also be found in the supplementary materials (Supplementary materials 2).

2.3. Laboratory analysis

DBS samples were processed using a modified elution protocol [29]. In Q2 2024 the samples were analyzed for IgG against tetanus, pertussis, and measles using three commercially available enzyme-linked immunosorbent assays (ELISA) (ESR108G, ESR1201G and ESR102G, respectively, from Serion, Wurzberg, Germany). Serum-equivalent concentrations were derived using a DBS-to-serum conversion factor using serum blood samples from adults in Belgium. Due to the absence of international standardization for measles binding assays and the variabilities associated with DBS use, two seropositivity thresholds (150 and 275 IU/L) were used, to capture a more nuanced picture of the population's immunity. Reporting results according to both thresholds allows for comparison with existing literature and acknowledges the uncertainty surrounding measles IgG correlates of protection. The protective 0.1 IU/mL threshold for tetanus was used, and titres >100 IU/mL for pertussis were considered indicative of recent vaccination or infection. More details on the laboratory analysis and cut-off justifications are provided in (Supplementary Materials 3).

2.4. Statistical analysis

The DBSs were analyzed for antibodies against measles, pertussis, and tetanus in both women and children. Using the sample size suggested by the WHO coverage survey provided the opportunity to assess vaccination coverage but not to draw associations with economic factors or compare clusters. Therefore, the results of the study are presented in a descriptive manner to report on coverage exclusively and without making an inferential analysis of socioeconomic factors as predictors of coverage. Data cleaning and descriptive analysis were performed in R version 4.3.1. The survey was analyzed using the classical WHO EPI cluster-survey method as described by Bennett et al. [30]. Because each cluster contributed approximately equal numbers of respondents, sampling weights were not applied; in this situation, the WHO EPI method estimates the overall proportion as the unweighted mean of the cluster-specific proportions. Ninety-five per cent confidence intervals were computed from the between-cluster variance of the cluster-level proportions. The design effect (DEFF) was defined as the ratio of the variance of the estimated proportion under the cluster design to the variance under simple random sampling and was calculated as $DEFF = \frac{s^2}{p^2(1-p)^2}$,

where s^2 is the between-cluster variance of the cluster-level proportions and p^- is the estimated overall proportion.

Concerning the maternal tetanus vaccination, women were classified as 'Fully Immunized' if they received the recommended number of tetanus vaccine doses for their total pregnancies: two doses for one pregnancy, three for two, four for three, or five for lifelong protection. Women who did not meet these thresholds were classified as 'Not Fully Immunized'. The women with no doses were labeled as 'Not Immunized'. While this classification does not account for the timing of doses, it allows to assess vaccination coverage in the region, focusing on current maternal and neonatal protection status rather than lifelong immunity.

2.5. Ethical considerations

The study protocol was approved by the DRC's National Health Ethics Committee (Approval No. 441/CNES/BN/PMMF/2023). Written informed consent was obtained from all participants, with consent documents provided in French and Lingala. Anonymized participant IDs were generated before data collection to maintain confidentiality during laboratory and data analysis procedures.

3. Results

Overall, DBSs were collected from 420 participants (210 women and 210 children) across 30 clusters (villages/neighborhoods) within Boende Health Zone. Due to missing or incomplete records in the survey database, we were able to obtain usable data pairs (survey response and DBS samples) for 413 participants (208 women and 205 children). Refusals to participate were recorded in 28 of the 30 clusters. Across these clusters, 71 eligible women or caregivers of eligible children declined participation before the target sample size was reached.

The majority of survey responders (70 % of the children caregivers and 80 % of women who recently gave birth) were younger than 34 years. Eight women and two caregivers were below the age of 18. Around half of the respondents had no formal education or some level of primary education. Table 1 shows demographic numbers and percentages for both categories of survey respondents. The numbers demonstrated by the table provide an idea of the remoteness and economic status of the study population. Only 9 % (95 %CI: 6–12) of participants had access to electricity and less than a third had a mobile phone, which is a reflection of the absence of cell network coverage beyond a 10 km radius around Boende, the lack of electrical infrastructure, and the overall economic constraints of the population.

3.1. Serology data

3.1.1. Women's serology

None of the 208 women had pertussis IgG titers exceeding the 100 IU/mL threshold. For measles, seropositivity rates were 55 % at the stringent threshold (≥ 275 IU/L) and 81 % at the relaxed threshold (≥ 150 IU/L). For tetanus, which is covered by a national antenatal vaccination program, approximately 73 % of the women who recently gave birth had antibodies above the 0.1 IU/mL protective threshold. Table 2 contains the percentages of seroprevalence above cut-off points for the three pathogens.

Given that the recommended number of tetanus vaccine doses varies with pregnancy history, participants were categorized into three groups based on their reported number of pregnancies and cumulative vaccine doses received. Table 3 presents the number of female participants in each immunization category out of the 208 total, the number of women who had protective antibodies, and the corresponding seroprotected percentage with 95 % confidence intervals. As observed in the table, the reported vaccination status did not consistently align with the observed seroprotection. For example, over half of the women who had reportedly

Table 1

Age, education level, and amenities ownership/access of study participants in Boende Health Zone, DR Congo (June 2023).

Category	Women who gave birth 0–11 months ago		Caregivers of children aged 12–23 months	
	Median = 25, IQR = 9, 25th – 75th percentile (Q1 = 22 – Q3 = 31)		Median = 27, IQR = 12 25th – 75th percentile (Q1 = 23 – Q3 = 35)	
Age (years)				
15–24	98	47 % (44 %–50 %)	63	31 % (28 %–34 %)
25–34	74	36 % (33 %–39 %)	78	38 % (35 %–41 %)
35–44	36	17 % (14 %–20 %)	41	20 % (18 %–22 %)
Over 44	–	–	9	4 % (3 %–5 %)
Missing value	–	–	14	7 % (5 %–8 %)
Total	208	100 %	205	100 %
Education level				
No education	44	21 % (17 %–25 %)	47	23 % (21 %–25 %)
Primary	60	29 % (26 %–32 %)	67	33 % (30 %–36 %)
Secondary	83	40 % (37 %–43 %)	66	32 % (29 %–35 %)
Tertiary	21	10 % (8 %–12 %)	16	8 % (5 %–11 %)
Missing value	–	–	9	4 % (3 %–6 %)
Total	208	100 %	205	100 %
Amenities ownership				
Access to electricity				
Yes	18	9 % (6 %–12 %)	18	9 % (7 %–10 %)
No	190	91 % (89 %–94 %)	187	91 % (89 %–94 %)
Cell phone ownership				
Yes	57	27 % (23 %–32 %)	39	19 % (15 %–23 %)
No	151	73 % (69 %–77 %)	166	81 % (77 %–85 %)
Radio ownership				
Yes	115	55 % (51 %–59 %)	115	56 % (53 %–59 %)
No	93	45 % (41 %–49 %)	90	44 % (41 %–47 %)
Motorbike ownership				
Yes	23	11 % (9 %–13 %)	14	7 % (6 %–8 %)
No	185	89 % (87 %–91 %)	191	93 % (92 %–95 %)

Table 2

Antibody levels above the predetermined cut-off points for measles, tetanus, and pertussis among 208 female participants who gave birth 0–11 months before sampling, with 95 % confidence intervals (CI) in Boende Health Zone, DR Congo (June 2023).

Pathogen	N	N above predefined cut-offs	DEFF	% (95 % CI)
Measles (275 IU/L)	208	114	0.12	55 % (53 %–57 %)
Measles (150 IU/L)	208	169	0.16	81 % (79 %–83 %)
Tetanus	208	151	0.22	73 % (70 %–75 %)
Pertussis	208	0	NA	0 %

Table 3

Immunization status and tetanus seroprotection among female participants who gave birth 0–11 months before sampling (N = 208) in Boende Health Zone, Tshuapa, DR Congo (June 2023).

Immunization status	N (% of 208)	Seroprotected ¹	DEFF	% (95 % CI)
Fully immunized	87 (42 %)	72	0.20	83 % (79 %–86 %)
Partially immunized	90 (43 %)	62	0.32	69 % (63 %–74 %)
Non-immunized	31 (15 %)	17	0.34	55 % (45 %–65 %)

¹ Seroprotected: having antibodies above the 0.1 IU/mL protection cut-off point.

never been vaccinated had antibodies above the 0.1 IU/mL protective threshold.

3.1.2. Children's serology

In terms of serology data, and among the 205 children in the survey, 10 [5 %, 95 %CI: (3 %–7 %)] had pertussis antibody levels exceeding 100 IU/mL, suggesting either recent exposure to the disease or recent vaccination. However, only one of these 10 children was reportedly fully vaccinated with three doses of the DTP + HepB+Hib vaccine (caregiver recall). For tetanus, which is administered in combination with pertussis in the pentavalent vaccine, 37 children [18 %, 95 %CI: (15 %–21 %)] had antibody levels above the protective threshold.

In terms of vaccination status, 108 children reportedly received the first dose of the pentavalent DTP + HepB+Hib vaccine, while 59 were reported to have received all three doses, with 12 of these having verifiable vaccination cards. A total of 89 children (43 %) were reportedly unvaccinated. Out of the 12 children with vaccination cards indicating receipt of three doses of the DTP + HepB+Hib vaccine, none had pertussis antibody levels above the predefined threshold, and only one had tetanus antibodies above the protective threshold. Table 4 presents the distribution of children categorized as vaccinated, partially vaccinated and unvaccinated, along with the number and percentage of those with antibodies against the three pathogens above the predetermined cut-off points.

For measles, antibodies above the two seropositivity cut-offs (275 IU/L and 150 IU/L) were present in 37 (18 %) and 62 (30 %) of children, respectively. Among all enrolled children (205), 127 (62 %) were reportedly vaccinated against measles, although only 9 of them (8 %) had vaccination cards as proof, while 78 children (38 %) were reportedly unvaccinated. Of those reportedly vaccinated, 23 children (18 %) had antibody levels above the 275 IU/L threshold, increasing to 40 children (32 %) when the cut-off was lowered to 150 IU/L. Interestingly, only one out of nine children with vaccination cards confirming measles vaccination had antibody levels above both cut-off points. Additionally, caregivers reported that 68 children had previously contracted measles, 27 (40 %) of these had antibody levels above the cut-off point of 275 IU/L. Median antibody titers were notably higher in children with a history of measles infection (102 IU/L, IQR: 916) compared to vaccinated children (16.5 IU/L, IQR: 171) and unvaccinated, non-infected children (16.5 IU/L, IQR: 48.9). Fig. 3 is a visual representation of the serology and vaccination status of children for all three pathogens.

In 11 out of 30 clusters, none of the surveyed children were (reportedly) fully vaccinated with three doses of the pentavalent vaccine (DTP + HepB+Hib). For measles, which is administered as a single dose and often supported by supplementary immunization activities, one cluster had all seven children reportedly unvaccinated. This pattern was mirrored in the serological results. In that cluster with no reportedly measles-vaccinated children, none of the seven children had antibody levels above the predefined cut-off, and similar gaps in immunity were observed in several other clusters.

Furthermore, when examining antibody levels independently of reported vaccination status, we found that none of the child participants had tetanus antibodies above the protective threshold in 10 out of 30 clusters. For measles, no children had antibody levels above the stringent cut-off of 275 IU/L in 11 clusters. Even when applying the more relaxed cut-off of 150 IU/L, 6 clusters still had no children with measles antibodies above 150 IU/L. Notably, using the higher measles cut-off (275 IU/L), five clusters had no children with antibody levels above the defined cut-offs for any of the three pathogens (measles, pertussis, or tetanus). Using the lower cut-off (150 IU/L), this number decreased to four clusters. Fig. 4 is a visualization of seroprevalence stratified by cluster.

4. Discussion

We conducted a study on vaccination coverage and seroprevalence of measles, tetanus, and pertussis antibodies in 12–23 months old children and women who were 0–11 months postpartum in 30 villages/neighborhoods (clusters) in Boende Health Zone in the DRC. A total of 71 participants refused to participate before attaining the target sample size of 420. This reluctance might be due to the study's request for blood samples (from children) -even in the form of a few drops- which could have deterred many people. The procedure may have been seen as inconvenient and unfamiliar, especially since blood collection is not a common practice in coverage surveys, particularly in remote settings outside of Boende town.

Measles antibodies above the 275 IU/L threshold were detected in 55 % (CI: 53 %–57 %) of the women in our sample. While vaccine-induced immunity naturally wanes with age, the decline is typically slower for live attenuated vaccines such as the measles vaccine. The observed level in our study is well below the 95 % threshold needed for herd immunity. The findings are not unexpected considering the likely

Table 4

Number (N) and percentage with 95 % confidence intervals for the level of seropositivity (SP) of 205 children (12–23 months) with antibodies above the cut-off point for pertussis, tetanus, and measles, stratified by vaccination status (fully vaccinated by card and caregiver recall, partially vaccinated, and unvaccinated) (June 2023).

	Vaccinated							Partially vaccinated (card and recall)				Unvaccinated ⁴ (card and recall)				Total			
	Card available			Caregiver's recall				N	SP	DEFF	%	N	SP	DEFF	%	N	SP	DEFF	%
Pertussis	12	0	0 %	47	1			57	3	3.47	5%(0 %–15 %)	89	6	0.75	7 %(2 %–11 %)	205	10	0.62	5 % (3 %–7 %)
Tetanus	12	1	8 % (– –)	47	10	0.38	21 % (14 %–28 %)	57	11	0.45	19 % (12 %–26 %)	89	15	0.50	17 % (11 %–22 %)	205	37	0.25	18 % (15 %–21 %)
Measles 275 IU/L	9	1	11 % (– –)	118	22	0.22	19 % (15 %–22 %)	NA ²	NA		NA	78	14	0.36	18 % (13 %–23 %)	205	37	0.16	18 % (16 %–20 %)
Measles 150 IU/L	9	1	11 % (– –)	118	39	0.36	33 % (28 %–38 %)	NA	NA		NA	78	22	0.29	28 % (23 %–34 %)	205	62	0.19	30 % (28 %–33 %)
Reported clinical measles ³	5	1	20 % (– –)	36	15	0.28	42 % (33 %–50 %)	NA	NA		NA	27	11	0.44	41 % (29 %–53 %)	68	27	0.42	40 % (32 %–47 %)

¹ SP in the table refers to “seropositive” which is defined here as the number of participants with antibody levels above the aforementioned cut-off points.

² As the measles vaccine was comprised of one dose only, the “partially vaccinated” fields were not applicable.

³ For the reported clinical measles, the more stringent cut-off of 275 IU/L was used given that natural infection with measles would mount high antibody response.

⁴ Unvaccinated means that the child did not receive any dose against the specific disease in the first column, and does not mean not vaccinated against any disease.

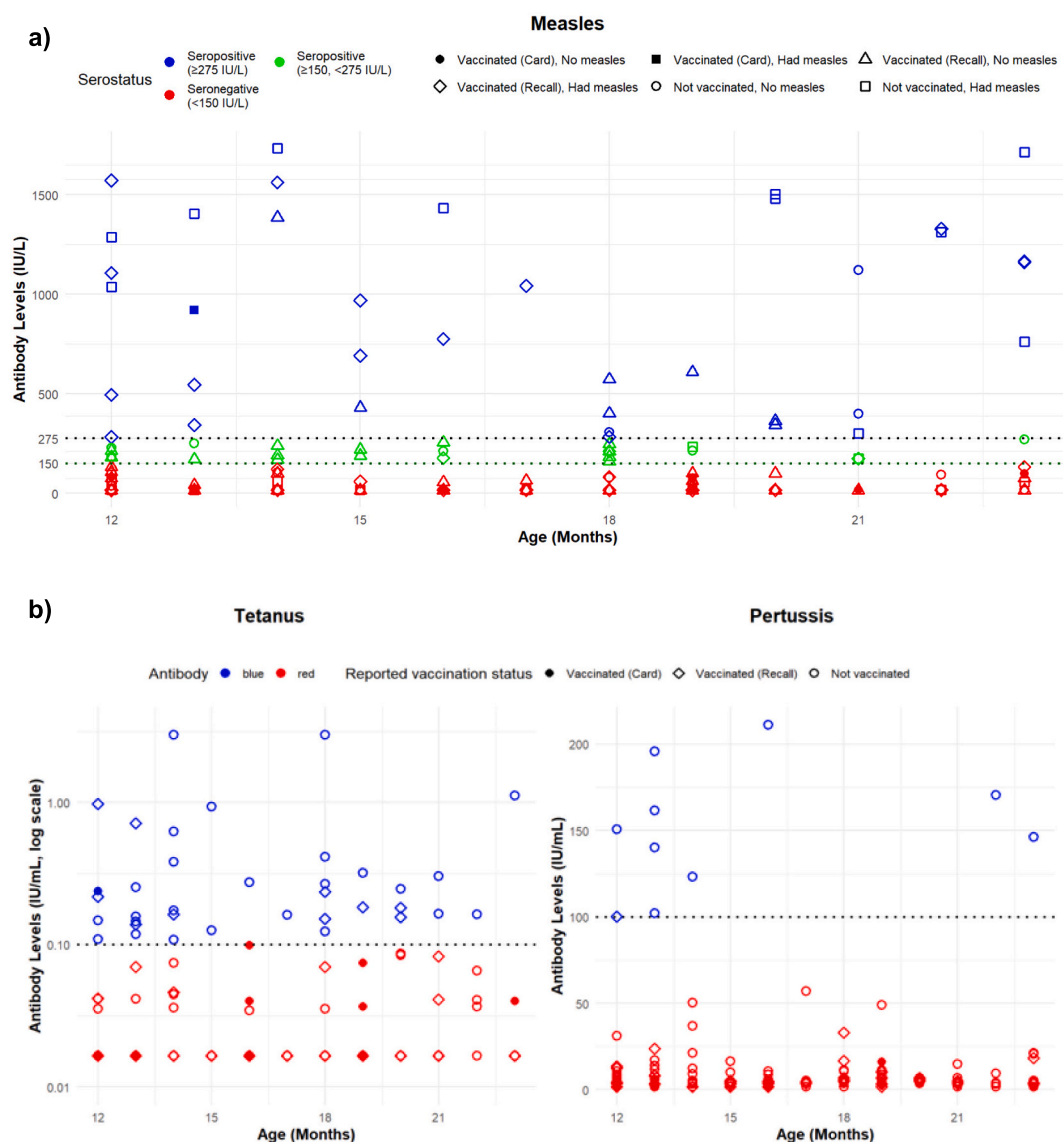


Fig. 3. Scatter plots showing antibody levels against measles, tetanus, and pertussis among children aged 12–23 months (June 2023). **a)** Measles antibody levels (IU/L), displayed with two cut-off points (150 and 275 IU/L), reported vaccination status, and self-reported history of measles infection. **b)** Tetanus and pertussis antibody levels, shown with their respective cut-off points and stratified by reported vaccination status (completed 3 doses of the pentavalent vaccine). For tetanus, blue points indicate antibody levels above the seroprotective threshold, while red points represent children with levels below seroprotection. For pertussis, blue points suggest recent vaccination or infection, whereas red points indicate the likely absence of recent vaccination or infection. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

low measles vaccination coverage during the childhood years of these women in Boende as well as the relatively early timing of the single-dose measles vaccine, which may have resulted in a suboptimal immune response due to immune system immaturity at that age.

Nevertheless, a review of 8 studies from the WHO African region put general population seroprevalence of measles antibodies above 92.5 % [31]. This is comparable to our data when using the less stringent cut-off point of 150 IU/L where the seropositivity for measles in postpartum women was assessed to be 81 %. Unlike with children, we could not compare antibody prevalence for measles and pertussis to the vaccine uptake against these disease for women, as adult women were naturally unable to recall which vaccines they had received in childhood.

In regards to the efficacy of the tetanus maternal vaccination program, the 2023–2024 DHS data suggested that at the national level, 57 % of women had received all recommended doses. In Tshuapa province, this percentage was 34 %, according to DHS [5]. The latter is close to the 42 % (95 % CI: 38 %–45 %) of efficiently vaccinated women (i.e. those

who received an appropriate number of doses according to the number of pregnancies) reported in our study. However, the percentage of women with antibodies above the protection threshold was higher than the number of women reportedly “fully immunized” in our sample [73 % (95 % CI: 70 %–75 %)], suggesting that either protection could be achieved with less than the recommended number of doses, or that women might have underestimated the number of doses received due to recall bias. In 2019, the country managed to eliminate neonatal tetanus [32]. However, our data shows a lower level of vaccination in the Boende health zone than the target of ≥ 90 % of newborns protected at birth, that many countries had set with the support of WHO [32,33]. Based on the findings of this study, the antenatal tetanus vaccination program will need to be improved and closely followed up in order for the country to keep its elimination status.

Serological analysis of 208 women revealed no pertussis antibodies above 100 IU/mL in any participant. This aligns with expectations given the absence of a maternal pertussis vaccination program in the country

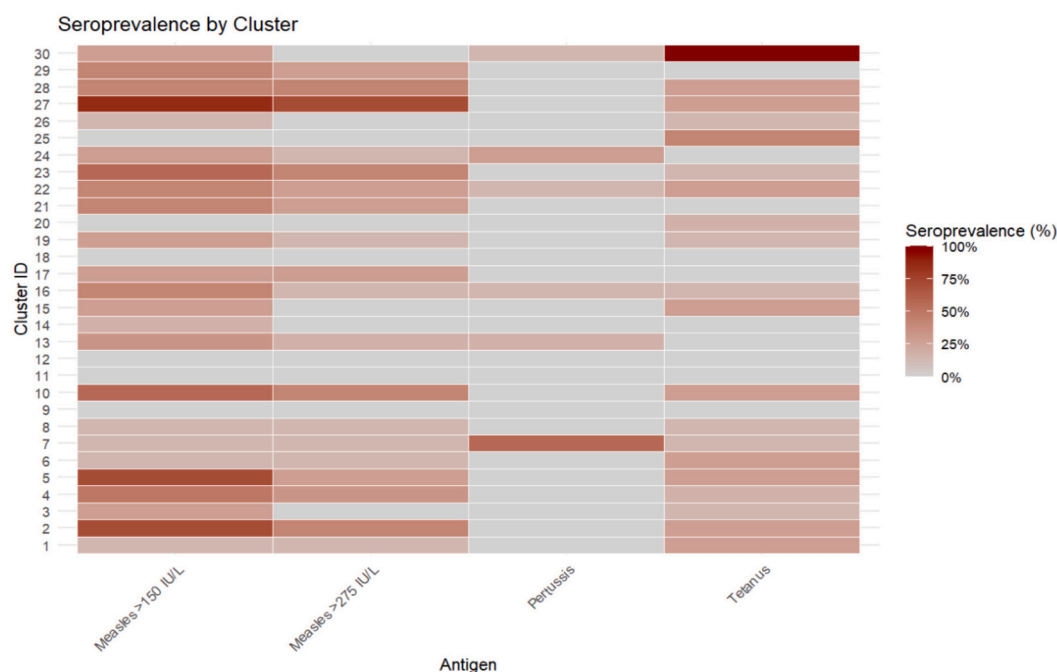


Fig. 4. Heat map of the number of children (0 to 7) who have antibodies against measles, pertussis, and tetanus above the defined cut-off points in each of the 30 clusters (June 2023).

and may also suggest low recent regional circulation of *Bordetella Pertussis*. However, the lack of antibodies suggests that no maternal antibody transfer is occurring. As a result, neonates and young infants born to these women are unlikely to receive passive protection against pertussis during the vulnerable period before their primary immunizations. Given that infants under one year of age are at the highest risk for severe pertussis and complications globally, and in light of the over 250,000 pertussis cases reported in the DRC in 2021, our findings strongly support the addition of maternal pertussis to the national immunization schedule to protect infants at birth [34,35]. This could be provided in the form of a trivalent Tdap vaccine with an acellular pertussis component as is the case in many countries around the world [36]. The benefits of adding the pertussis vaccine will become more apparent if it is coupled with efforts to raise maternal antenatal vaccination coverage in the country.

Among children who were 12–23 months of age, we observed low seroprevalence of antibodies against all three diseases assessed. Especially the 10 (5 %) of children who had pertussis antibodies above the 100 IU/mL cut-off point was strikingly low. While no correlate of protection exists for pertussis, it is established that pertussis antibodies wane quickly. For example, a study of children from different African countries found that only 29.5 % of children vaccinated against pertussis had anti-PT antibodies at age 18–36 months [37]. Therefore, the absence of antibodies above the cut-off point does not mean that a child has never been vaccinated or has not encountered the disease, but it refers to the absence of a recent infection or recent vaccination [38]. This might explain why 8 of the 10 observations of high antibodies were detected in children 16 months of age and younger. However, the selected age range in the current study is relatively narrow (12 months) and close to the typical vaccination age. Given this, along with the overall low seroprevalence of tetanus antibodies (18 %), we suspect that low vaccination coverage, rather than antibody waning, is the primary factor behind this finding. This observation is supported by the 2023–2024 DHS data, where only 7.7 % of the children population in Tshuapa were reported to be vaccinated with three doses of the pentavalent vaccine (DTP + HepB+Hib) [13].

In our study, only 18 % (95 % CI: 16 %–20 %) of children aged

12–23 months had measles antibody concentrations above 275 IU/L, a threshold that has been associated with a high likelihood of neutralizing immunity [39]. Even when applying a less stringent cut-off of 150 IU/L, the seropositivity rate remained low at 30 %. These findings may partially explain the recurrent measles outbreaks and continued disease burden in the country [12,40]. Surveillance data indicate that measles outbreaks have continued to occur across several provinces of the DRC, affecting both vaccinated and unvaccinated children. In Tshuapa province, where our study took place, more than 13,000 suspected cases and nearly 400 deaths were reported between 2018 and 2020, corresponding to a cumulative incidence of around 6000 per million inhabitants [41]. Although data for 2021–2023 are limited at the provincial level, cases continued to be reported in Tshuapa, Equateur, and other provinces, but in smaller numbers compared to the 2019–2020 period, suggesting that transmission remained active in the wider region [42,43]. The persistence of outbreaks in 2024 in Tshuapa province (over 5000 suspected cases in 2024) [44] despite large-scale vaccination campaigns points to continued immunity gaps, consistent with the low antibody concentrations observed in our cohort.

A study of four other health zones in the southeast of the DRC reported measles seropositivity for children up to 59 months was higher ranging between 49 % and 93 %, and so was (reported) vaccination coverage (75 %–85 %) [45]. The higher rates are naturally expected given the larger age range (up to 5 years of age) which gives more chance for children to be vaccinated, or contract the infection. Additionally, the DHS data for 2023–2024 shows that measles vaccination coverage in provinces of these health zones (Haut-Lomami, Lualaba, Tanganyika) are all (slightly) higher (46 %, 58 %, and 40 % respectively) than the measles vaccination coverage reported in Tshuapa (30 %).

Importantly, antibody levels below a cut-off in an ELISA assay do not necessarily indicate a lack of vaccination. The CDC estimates primary vaccine failure (the inability to mount an antibody response following vaccination) to occur in 2 %–7 % of children when the measles vaccine is administered as a single dose at or after 12 months of age [46]. However, in the DRC, the first dose is administered at nine months, a policy aligned with WHO guidelines for high-risk countries, where early protection is essential due to the high burden of the disease [47]. That said,

Immunization at nine months is associated with lower seroconversion rates (85 % at 9 months vs. 90 %–95 % at 12) due to multiple factors, including the potential effect of immune system immaturity and residual maternal antibodies -though expected to be minimal at nine months of age- [47,48]. Recognizing this limitation, the DRC Ministry of Health introduced a second dose recommendation at 15 months of age, particularly following the analysis of 2013–2014 DHS data [49]. Nevertheless, this was not yet implemented in Tshuapa for the cohort included in our study, but was implemented later that year (2023) according to local health workers.

Among children who are vaccinated according to vaccination cards, the proportion who had antibody levels above the cut-off point was low for all three pathogens: 0 % (0 of 12) vaccinated against DTP + HepB+Hib had pertussis titres above the cut-off point, 8 % (1 of 12) had protective tetanus antibodies, and 11 % (1 of 9) vaccinated against measles were seropositive (Table 6). Although the number of children vaccinated according to vaccination cards was small, the fact that most of these children had antibody levels below the cut-off points is concerning. This exceeds what would typically be expected from vaccine failure or waning immunity. Therefore, this discrepancy may point to potential issues with the accuracy of vaccination records, including the possibility of falsified or incorrectly completed cards. However, such findings must be interpreted with caution and warrant further investigation by trained monitors through robust data quality assessments.

Moreover, the absence of antibody prevalence across entire clusters could suggest systemic issues with vaccination coverage, potentially indicating that certain cohorts in specific locations lack access to vaccination. On the other hand, it could be due to widespread vaccine hesitancy. Even beyond these clusters with zero antibodies, vaccine hesitancy could be a main driver for low vaccine uptake. A study published in 2023 reported that “82.15% of the parents/guardians of the ‘zero dose children’ in DRC said they wanted their children to receive ‘none’ of the recommended vaccines” [50].

While the lack of vaccination coverage probably constitutes the main reason for the low seroprevalence of antibodies against different vaccine-preventable infections, many other explanations could have contributed to the observed results. For instance, measles-associated immune amnesia resulting from the loss of pre-existing immune memory, may impair the ability to boost antibody responses upon re-exposure or vaccination [51]. This reduced boosting capacity could contribute to lower antibody concentrations observed in this cohort, especially following the large measles outbreak in 2023. A study in 2013–2014 in the DRC found that children with a history of measles infection were less likely to have tetanus antibodies after three vaccine doses compared to children who did not contract a measles infection [18]. Another reason for low seroprevalence could be the potency of the vaccines in the absence of a reliable cold chain. This includes the storing and transporting conditions at the provincial levels where fridges, freezers, and ice boxes are sometimes not available or temporarily not functional. For example, many measles-containing vaccines are stable for a maximum of a week out of the cold chain and lose potency afterwards [52]. Additionally, these vaccines become less potent within an hour after the reconstitution of multidose lyophilized vials, if the vial was opened and not preserved between 2 and 8 degrees Celsius [53]. In Boende, a qualitative study conducted by our team showed that this is a concern for the provincial EPI chief medical officer and several town inhabitants and that it affects vaccine confidence in the area [54]. Other factors that could contribute to low seroprevalence include the persistence of maternal antibodies (mainly applicable for vaccines given early in infancy like DTP + HepB+Hib), which can inhibit the generation of infant antibodies against vaccines that are administered before the first birthday, as well as co-infections with HIV or helminth infections at the time of vaccination, which are known to suppress immune responses or affect the immune response to vaccines [55,56]. Additionally, vitamin A deficiency may impair immune responses, though these factors were not directly assessed in our study [57]. Past malaria episodes could be

another reason for hampered immune responses against vaccines given the endemicity of the parasite in the province/country [58,59]. While these additional factors may have contributed to the observed results, we believe that the primary driver for low seroprevalence appears to be the low vaccination coverage in this region, as supported by DHS data and the qualitative findings highlighting systemic challenges in vaccine delivery.

Strengthening the EPI program and surveillance is a task always present on the agendas of different (non) governmental stakeholders. We believe that this will be even more urgent and necessary as the malaria vaccine rollout starts [60]. Perhaps this vaccine rollout will help raise vaccine confidence, as malaria is a widely feared disease and caregivers could be more willing to go to health centers to vaccinate children against malaria than against other diseases perceived as less immediate threats [61]. As the malaria vaccine is given in four doses, this rollout could offer additional occasions to give other vaccines in parallel.

5. Limitations

Interpreting the data presented in this study requires consideration of several limitations related to both data collection and analysis. First, using the 2019 population estimates for a 2023 study may not fully capture demographic shifts, but they remained the most reliable available source. Second, following a more recent version of the WHO vaccination coverage cluster survey tool (WHO/IVB/18.09) might have improved the statistical accuracy of coverage estimates. Third, vaccination histories were mainly based on caregiver recall rather than vaccination cards, which prevented assessment of timeliness and may have introduced recall or desirability bias. Indeed, the proportion of reportedly vaccinated children was higher than the number of children with protective antibody levels, as well as higher than figures from the DHS. Additional training for field staff could help reduce such biases in future surveys. Lastly, refusal to participate was encountered in some cases, which may have overestimated the coverage and seroprevalence results (if the non-responders were assumed to be less vaccinated). However, this is expected in surveys involving blood sampling, as cluster sampling allowed for continuation until the target sample size was reached, minimising the impact of non-response on coverage and seroprevalence estimates.

Beyond survey design, laboratory-related factors may also have contributed to the discrepancy between reported vaccination status and antibody titers. For example, DBS sample quality may have been compromised by hemolysis during collection or degradation through transportation, particularly between remote locations and Boende hospital. Additionally, no positive control samples from the field were available to assess assay performance or validate the DBS-to-serum conversion factor under local conditions. Instead, the conversion factor was derived from paired serum and DBS samples from healthy adults in Belgium, which may not fully reflect the immunological or physiological profiles of children in the DRC. These factors could have influenced the observed antibody levels. Therefore, we cannot entirely rule out that technical limitations may have contributed to the low antibody levels observed. Future studies should incorporate external controls from the same population to better validate serological findings.

Finally, although Plaque Reduction Neutralization Tests (PRNT) are the gold standard for measuring functional measles immunity, we used ELISA-based IgG detection due to logistical constraints. While ELISA is widely used in large-scale studies, it lacks international standardization and a universally accepted cut-off for measles seroprotection [45,49]. Future work should focus on locally validating measles IgG assays on DBS in the DRC. Despite these limitations, we believe our study offers valuable insights into vaccination coverage and immunity gaps in a remote region, highlighting the need for improved vaccination documentation, serosurveillance, and service delivery.

6. Conclusion

This study identified low seroprevalence of antibodies against three vaccine-preventable infections, measles, tetanus, and pertussis, among children aged 12–23 months in Boende Health Zone in the DRC. These findings indicated a low childhood vaccination coverage in the study area and provided plausible explanations for the recurring outbreaks of different diseases, particularly measles. The low measles seroprevalence observed in Boende, despite national outbreaks in 2022 and 2023, may point to the geographically uneven nature of measles transmission, underscoring the importance of localised surveillance data in guiding targeted vaccination efforts. The serology component revealed clear gaps in the immunization program, especially in settings where no vaccination card was available. Additionally, incorporating serology into coverage studies could provide independent validation of vaccination records, allowing for the detection of inaccuracies and helping monitor the performance and credibility of immunization reporting systems. As for maternal vaccination, the absence of pertussis antibodies above the cut-off point in the entire adult female study population highlights the need for the inclusion of the vaccine as part of antenatal care, especially given the global rise in the number of pertussis cases and the potential serious effects on young children. Continuing to do (sero) surveillance in different parts of the country, especially in remote areas, alongside improving logistic access to vaccines, is critical to identify gaps and enhance immunity levels. This is particularly important in light of recent cuts to international aid funding, such as reductions in the United States' funding, which have historically supported large-scale surveys like the DHS. Future research and surveillance efforts should also include cold chain components to ensure that potent vaccines are being administered. Together, these actions will strengthen the resilience of the national immunization program and contribute to sustained improvements in population immunity.

CRedit authorship contribution statement

Maha Salloum: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Leen Vandermosten:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Tafadzwa Maseko:** Writing – review & editing, Software, Formal analysis. **Solange Milolo:** Project administration, Investigation. **Ynke Lariviere:** Writing – review & editing. **Trésor Zola Matuvanga:** Writing – review & editing. **Freddy Bikioli Bolombo:** Methodology. **Vivi Maketa Tevuzula:** Writing – review & editing. **Hypolite Muhindo-Mavoko:** Writing – review & editing. **Antea Paviotti:** Writing – review & editing, Supervision. **Hilde Bastiaens:** Writing – review & editing. **Pierre Van Damme:** Writing – review & editing. **Isabelle Desombere:** Writing – review & editing, Methodology. **Jean-Pierre Van Geertruyden:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Patrick Mitashi:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT in order to enhance the flow and readability of the text. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Tresor Zola Matuvanga reports a relationship with Essential Program of Immunization (EPI) that includes: employment. If there are other authors, they declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.vaccine.2025.128083>.

Data availability

Data will be made available on request.

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